NEERAV JAIN

Applied Machine Learning Engineer & Systems Researcher

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EDUCATION

ABES Engineering College • B.Tech in CSE (Artificial Intelligence & Machine Learning) **2025 – 2029**
Relevant Focus: Deep Learning, Computer Vision, Spatiotemporal Modeling, Systems Architecture *Ghaziabad, UP, India*

Government Polytechnic • Diploma in Electronics Engineering **2022 – 2025**
Foundational training in embedded microcontrollers, ADC sensor signal conditioning, and hardware interfacing *Ghaziabad, UP, India*

INDUSTRIAL EXPERIENCE

Cognifyz IT Solutions Pvt. Ltd. • Front-end Development Intern **Apr 2026**
- Built responsive, high-performance web interfaces and interactive data intelligence dashboard modules in an agile team setup.
  - Integrated asynchronous REST endpoints with zero layout shift, maintaining cross-browser compatibility and optimized mobile web rendering.

Rajendra Gears • IoT Engineering Trainee **Aug 2024 – Sep 2024**
- Engineered and deployed 3 production-ready IoT telemetry prototypes integrating NodeMCU (ESP8266), Raspberry Pi 4, and Arduino into live factory machinery.
  - Reduced manual machinery monitoring effort by ~40% and achieved sub-200 ms response latency in real-time sensor acquisition and threshold-based alarms.

KEY ENGINEERING & RESEARCH PROJECTS

AeroPulse-X: Mission-Aware AI Digital Twin for MALE-UAV Engine Health **2026**
Smart India Hackathon (SIH26054) • PyTorch, TCN, HistGradientBoosting, FastAPI, Three.js
- Architected a real-time propulsion health digital twin for turbocharged Rotax 914 aero-piston engines combining reduced-order thermodynamic cycle simulation, ISA atmospheric coupling, and cross-sensor peer trust veto.
  - Engineered hybrid diagnostic fusion (HistGradientBoosting 0.70 + TCN 0.30) achieving **99.81% accuracy** on synthetic ODE trajectories and **89.47% accuracy (95.68% fault recall)** on real NASA ACES Altus II flight telemetry; certified by **308 passed pytest tests**.

North Indian Ocean Subsurface Temperature Reconstruction **2026**
Deep Learning Research • PyTorch 2.0+, xarray, netCDF4, GeoPandas, FastAPI, Vercel
- Reconstructed 15 depth-resolved vertical potential temperature profiles (0–1000m) from 7 satellite surface streams (OSTIA SST, SMAP SSS, CMEMS SSH, OSCAR currents, CCMP winds) using causal 5-day × 3×3 spatiotemporal receptive fields.
  - B8 Spatiotemporal Embedding Network achieved a **0.9800 °C column test RMSE** on an independent chronological holdout (Days 313–365 of 2020) with a 10-day purge buffer, representing a **22.11% error reduction over daily climatology (B1)**; verified with **106 passed tests**.

Event-Centric Multimodal Wildfire Intelligence for India **2026**
Authoritative Major System • Python 3.11, LightGBM, NASA FIRMS, ERA5-Land, NOAA ETOPO DEM
- Integrated 7 years of NASA FIRMS VIIRS 375m active fire telemetry with Copernicus ERA5-Land multi-timescale reanalysis and authoritative NOAA ETOPO 2022 digital elevation geomorphology (slope, aspect, elevation, ruggedness).
  - Evaluated under strict **Leave-One-Geographic-Regime-Out (LOGRO)** spatial cross-validation across diverse Indian eco-zones, achieving **0.8920 ROC-AUC and 0.7410 PR-AUC**; validated by **60 automated pytest tests**.

SYNAPSE (SIGNATURE VMAKE): Enterprise Biometric Signature Verification **2026**
Bank Muscat BRD Series • PyTorch, ResNet-18 Siamese Metric Learning, FastAPI, PostgreSQL
- Implemented writer-independent biometric signature verification platform using shared-weight ResNet-18 Siamese Deep Metric Learning and ForgeryAwareMetricLoss, coupled with multi-reference customer gallery centroid aggregation.
  - Demonstrated zero data leakage on unseen open-set signers (CEDAR held-out Writers 46–55), achieving **4.82% Equal Error Rate (EER) and 0.9840 ROC-AUC** with a complete 3NF audit database; certified with **53 passed tests**.

Bank Statement Table Extraction & Banking Analytics **2026**
Enterprise Banking Deliverable • FastAPI, PyMuPDF (fitz), scikit-learn, DistilBERT NLI, Pydantic v2
- Developed coordinate geometry token extraction and RandomForest line role classification pipeline parsing multi-column banking statement PDFs with native 3-decimal OMR currency support and zero-shot DistilBERT transaction categorization.
  - Achieved **99.10% transaction table extraction recall** and **100% running balance accounting consistency** across **41 automated unit and security tests**.

TECHNICAL TOOLING & SKILLS

Machine Learning & Statistics:	Scikit-learn, HistGradientBoosting, LightGBM, Platt Scaling Calibration, Isolation Forest, Hard-Negative Mining, Metric Learning, Chronological Purge Buffering, LOGRO Spatial Cross-Validation
Deep Learning & Architectures:	PyTorch 2.0+, Temporal Convolutional Networks (TCN), Siamese Metric Networks (ResNet-18), Causal GRU/BiGRU, Multi-Head Attention, Hugging Face Transformers, OpenCV, YOLO
Earth Observation & Geospatial:	NASA FIRMS VIIRS 375m, Copernicus ERA5-Land, NOAA ETOPO 2022 DEM (SRTM v3), Copernicus GLORYS12V1, OSTIA SST, Argo CTD Float Calibration, xarray, netCDF4, GeoPandas
Full-Stack & Systems Engineering:	Next.js 14 (App Router), TypeScript, React 18, FastAPI, Uvicorn, Python, Three.js WebGL, Tailwind CSS, PostgreSQL, SQLite (3NF Schemas), Pydantic v2, REST APIs
DevOps, Testing & Hardware:	Pytest (500+ Automated Tests across 5 repos), Docker, Git, Vercel Fluid Compute, NodeMCU (ESP8266), Raspberry Pi 4, Arduino, Embedded C, UART/Serial Protocols